B2B SaaS Analytics Project Master Blueprint

Complete Portfolio Guide for MySQL (DBeaver) & Power BI Desktop | Version 3.0 (Plain Language Edition)

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1. Project Executive Overview

This blueprint provides a clear, step-by-step roadmap for building a complete B2B SaaS marketing, sales, and product analytics project. You will work with **11 connected tables** containing over 400,000 rows of realistic business data spanning 3 full years (2023–2025). The goal is to clean dirty raw data in MySQL (using DBeaver) and then build an interactive reporting dashboard in Power BI Desktop.

Summary of Dataset Improvements (V3.0 Update):
• **Realistic Company Names:** In dim_accounts, generic company placeholders have been replaced with realistic business names (e.g., *Nova Software LLC, Velocity Cloud Inc., Ironclad Systems Ltd.*).
• **Expanded Industries & Company Sizes:** Industries are expanded to **18 realistic business sectors** (Fintech, HRTech, HealthTech, etc.), and company sizes span **8 tiers** (1-10 through 10,000+ employees).
• **Geography Matching:** Countries span **23 global markets**. Each company's home country matches the geographical sales region of their assigned sales rep in dim_sales_reps, and fact_user_sessions countries match 100% with the company's home country.
• **Real Sales Rep Names:** dim_sales_reps now features real human first and last names (e.g., *Alexander Wright, Sophia Martinez, Marcus Chen*).
• **Natural Ad Target Keywords:** dim_ad_groups now includes human-like search queries (e.g., *["b2b saas platform pricing", "enterprise software reviews 2024"]*).
• **Top-of-Funnel Visitor Tracking:** In fact_web_app_events, missing account_id values (~30%) are **intentional by design**. This reflects real-life website tracking where anonymous visitors browse public web pages before signing up or logging in.

2. Marketing, Sales & SaaS Metrics Glossary

Category	Term	Definition & Easy Business Explanation
Marketing	**UTM Parameters**	Small tracking tags added to the end of website links (e.g., *utm_source=google*) so you can tell exactly where visitors and ad clicks came from.
Marketing	**MQL (Marketing Qualified Lead)**	A potential customer who has shown clear interest in your product (like visiting the pricing page or filling out a form) and is ready for the sales team to contact.
Marketing	**CAC (Customer Acquisition Cost)**	**Formula:** Total Sales & Marketing Cost / New Customers Signed Up Tells you how much money your company spends on marketing and sales on average to win one new paying customer.
Marketing	**ROAS (Return on Ad Spend)**	**Formula:** Revenue Made From Ads / Total Ad Spend Tells you how many dollars in revenue you get back for every single dollar spent on online advertising.
Marketing	**Attribution Modeling**	Rules that decide which marketing channel gets credit for a sale. • **First-Touch:** Gives 100% credit to the first ad or link clicked. • **Last-Touch:** Gives 100% credit to the final link clicked before buying. • **Linear:** Splits the credit equally among all ads or links clicked.
Sales	**SQL (Sales Qualified Lead)**	A lead that the sales team has spoken with and confirmed has the budget, authority, and need to actually purchase the software.
Sales	**Opportunity / Pipeline**	An active deal currently being negotiated with a customer, with an estimated dollar value attached.
Sales	**Win Rate %**	**Formula:** (Deals Won / Total Deals Negotiated) × 100 The percentage of active sales opportunities that successfully turn into paying customers.
Sales	**Sales Velocity**	**Formula:** (Deals × Avg Deal Value × Win Rate %) / Avg Days to Close Measures how fast money moves through your sales pipeline per day.
SaaS Metrics	**MRR & ARR**	**MRR:** Monthly Recurring Revenue — predictable monthly money from active subscriptions. **ARR:** Annual Recurring Revenue = MRR × 12.
SaaS Metrics	**Freemium Model**	A business model where anyone can use basic features for free, but must upgrade to a paid plan to unlock advanced tools or extra user seats.
SaaS Metrics	**Churn Rate %**	**Formula:** (Cancelled Subscriptions / Total Active Subscriptions) × 100 The percentage of paying customers who cancel their subscription during a period.
SaaS Metrics	**LTV (Lifetime Value)**	**Formula:** Avg Monthly Revenue Per User / Monthly Churn Rate % The total amount of money a customer is expected to pay you before they eventually cancel.
SaaS Metrics	**NRR (Net Revenue Retention)**	Measures whether revenue from existing customers is growing or shrinking over time, factoring in plan upgrades, downgrades, and cancellations.

3. Dataset Architecture (11 Tables)

Type	Table Name	Exact Rows	Primary / Foreign Keys	Simple Explanation & What It Contains
Dimension	**dim_product_plans**	4	PK: plan_id	Subscription plan options (Free, Starter Pro, Growth, Enterprise) with seat limits and prices.
Dimension	**dim_sales_reps**	25	PK: rep_id	List of sales reps with realistic full names, teams (Inbound/Outbound), assigned region, and sales targets.
Dimension	**dim_accounts**	5,000	PK: account_id FK: account_owner_id	Real-world B2B company names, 18 distinct industries, 8 company size tiers, signup dates, and home countries aligned with their sales rep's region.
Dimension	**dim_utm_channel_mapping**	8	Composite: utm_source, utm_medium	Maps raw web tracking links (UTM tags) to clean marketing channel names (e.g., Paid Search, Organic, Email).
Dimension	**dim_ad_groups**	60	PK: ad_group_id	Ad campaign structure with human-like target keywords (e.g., ["b2b saas platform pricing", "enterprise software reviews"]).
Fact	**fact_ad_campaign_spend**	16,440	PK: spend_id FK: ad_group_id	Daily ad dollars spent, ad impressions, and click counts across Google, LinkedIn, Meta, Twitter, and Bing over 3 years.
Fact	**fact_crm_leads_opportunities**	30,000	PK: lead_id FK: account_id, assigned_rep_id	Tracks sales pipeline progress (Lead -> MQL -> SQL -> Opportunity -> Closed Deal) with dates and deal amounts.
Fact	**fact_subscription_billing**	35,000	PK: subscription_id FK: account_id, plan_id	Recurring billing payments, monthly revenue (MRR), payment methods (Stripe, PayPal, Wire), and renewal dates.
Fact	**fact_subscription_churn_cancellations**	7,000+	PK: churn_id FK: subscription_id, account_id, plan_id	Detailed logs of cancelled accounts, exact cancellation dates, lost MRR, and customer exit feedback reasons.
Fact	**fact_user_sessions**	60,000	PK: session_id FK: lead_id, account_id	Web/App user visits (Web, Android, iOS), session durations, page view counts, and visitor country (100% matched to account home country).
Fact	**fact_web_app_events**	103,000	PK: event_id FK: session_id, account_id	User activity actions (clicks, signups, exports) with JSON parameters. ~30% missing account_id is intentional (anonymous pre-signup visitors). Contains 3,000 duplicate click logs.

4. Data Quality Flaw Index & Cleaning Defect Map

Defect Type	Affected Table(s)	Injected Defect Description / Example	Easy How-To Fix in MySQL
1. Mixed Date Formats	stg_dim_accounts stg_fact_crm_leads_opportunities stg_fact_subscription_billing stg_fact_user_sessions stg_fact_web_app_events	Dates are stored in different text formats: ISO (2024-03-15 14:22:10), US (03/15/2024 02:22:10 PM), and Unix Epoch numbers (1710512530).	Use CASE statements with STR_TO_DATE() and FROM_UNIXTIME() in SQL to convert all date columns into clean standard DATETIME.
2. Duplicate Event Logs	stg_fact_web_app_events	About 3,000 double-click event logs caused by website tracking tag glitches.	Use an SQL CTE with a window function (ROW_NUMBER() OVER (PARTITION BY session_id, event_name, event_timestamp ORDER BY event_id)) to keep only row #1 and filter out duplicates.
3. Semi-Structured JSON	stg_fact_web_app_events	Marketing tracking tags are trapped inside a single JSON text block (e.g., {"utm_source":"google","utm_medium":"cpc"}).	Use MySQL's JSON_UNQUOTE(JSON_EXTRACT(event_params_json, '$.utm_source')) to pull the values out into clean, separate SQL columns.
4. Inconsistent Keys & Extra Spaces	stg_dim_accounts	Some Account IDs are lowercase or formatted differently (ACC-00045 vs acc_45). Some company names have accidental spaces around them.	Use TRIM() to remove extra spaces and UPPER() or REPLACE() to make all IDs look identical.
5. Missing / Orphan Keys	stg_dim_accounts stg_fact_crm_leads_opportunities	Some accounts refer to sales rep IDs that do not exist (e.g., REP-028). Some leads refer to missing account IDs (e.g., ACC-99999).	Find these missing links using LEFT JOIN and replace missing values with 'Unassigned' or 'Unknown Account' using COALESCE().
6. Inverted Dates	stg_fact_crm_leads_opportunities	About 1% of sales deals show a closed_date that happens BEFORE the deal was created (created_at).	Use a CASE statement to catch these impossible date order errors and swap them back into logical order.

5. Master Project Roadmap & Execution Checklist

Phase & Stage	Action Item / Task	Status
Phase 1: Ingestion	Create saas_analytics_staging database in DBeaver.	**COMPLETED**
Phase 1: Ingestion	Enable local_infile = ON in MySQL and allowLoadLocalInfile = true in DBeaver driver properties.	**COMPLETED**
Phase 1: Ingestion	Execute DDL and LOAD DATA LOCAL INFILE script to populate all 11 staging tables.	**COMPLETED**
Phase 1: Ingestion	Verify staging table row counts (103k events, 60k sessions, 35k billing, 30k leads, 16.4k ad spend, 5k accounts, 60 ad groups, 25 reps, etc.).	**COMPLETED**
Phase 2: SQL Cleaning	Clean text fields: Remove extra spaces & standardize account ID casing in stg_dim_accounts.	**NEXT STEP**
Phase 2: SQL Cleaning	Clean timestamps: Convert all ISO, US, and Unix Epoch date strings into standard MySQL DATETIME format.	PENDING
Phase 2: SQL Cleaning	Extract JSON parameters: Pull utm_source, utm_medium, and utm_campaign out of event_params_json.	PENDING
Phase 2: SQL Cleaning	Remove duplicates: Write an SQL query using ROW_NUMBER() to remove the ~3,000 duplicate web event logs.	PENDING
Phase 2: SQL Cleaning	Fix orphan keys: Replace non-existent sales rep and account IDs with 'Unassigned' or 'Unknown'.	PENDING
Phase 2: SQL Cleaning	Create clean SQL Views: Save your cleaned tables as production SQL VIEWS (e.g., vw_fact_web_events).	PENDING
Phase 2: SQL Cleaning	Add Indexes: Create B-Tree INDEX on key columns (`account_id`, `session_id`, `lead_id`, `event_timestamp`) to speed up queries.	PENDING
Phase 3: Power BI	Connect Power BI Desktop to MySQL database via Import Mode using your clean production SQL views.	PENDING
Phase 3: Power BI	Build Star Schema: Connect fact tables to dimension tables using 1-to-Many single-directional relationships.	PENDING
Phase 3: Power BI	Create Date Table: Build a master Dim_Date table using DAX CALENDARAUTO().	PENDING
Phase 3: Power BI	Handle multiple dates: Link Dim_Date to `Created_Date`, `MQL_Date`, and `Closed_Date` using `USERELATIONSHIP()`.	PENDING
Phase 3: Power BI	Write DAX measures: Build measures for MRR, ARR, Active Subscribers, Churn Rate %, CAC, ROAS, and Win Rate %.	PENDING
Phase 3: Power BI	Marketing Attribution DAX: Build DAX measures to compare First-Touch vs Last-Touch revenue attribution.	PENDING
Phase 4: Refresh	Check Query Folding: Ensure Power Query steps get pushed back to MySQL so the database engine does the heavy lifting.	PENDING
Phase 4: Refresh	Set up Incremental Refresh: Create RangeStart and RangeEnd Date/Time parameters in Power Query.	PENDING
Phase 4: Refresh	Configure Refresh Policy: Set Power BI to keep 3 years of historical data and refresh only the last 30 days.	PENDING

6. Business Questions & Portfolio Challenges

- **UTM Channel Performance:** Write an SQL query extracting utm_source from JSON in stg_fact_web_app_events, join with stg_dim_utm_channel_mapping, and calculate total events per marketing channel.
- **Lead Velocity & Bottlenecks:** Calculate average time (in days) a lead spends between stages (Lead → MQL → SQL → Opportunity → Closed Won) using SQL window functions (DATEDIFF, LEAD).
- **Subscription Cohort Churn:** Group subscribers by signup month and calculate 30-day, 60-day, and 90-day subscription churn rates in SQL.
- **Sales Rep Quota Attainment Stored Procedure:** Write a MySQL stored procedure accepting Quarter and Year parameters that returns each sales rep's closed-won deal volume, quota target, and attainment %.
- **Executive SaaS Power BI Dashboard:** Build an executive dashboard displaying MRR, ARR, Active Subscribers, CAC, and Net Churn Rate % over 3 years with YoY growth time intelligence.
- **Marketing ROAS & Attribution Explorer:** Create a visual report comparing campaign spend against generated pipeline revenue across Google, LinkedIn, and Meta ads.
- **Product Engagement vs Churn Risk:** Build a scatter plot / cohort matrix in Power BI comparing session duration and feature usage events against customer churn probability.